🎢 Theme Park · One Function, Many Coasters — Parameters — English Worksheet

Topic: One Function, Many Coasters — Parameters · **Course:** Theme Park Creator · **Time:** about 45 minutes

This lesson you build a roller coaster with a **function**. You give it **parameters** — numbers it uses inside — and change them to build a long coaster, a tall spiral, or a wide one.

These are the blocks you use this lesson:

```
rollerCoasterBuilder.add_straight_line(10)
rollerCoasterBuilder.add_ramp(RcbVerticalDirection.UP, 10)
rollerCoasterBuilder.add_spiral(RcbVerticalDirection.UP, TurnDirection.LEFT, 10, 3)
rollerCoasterBuilder.add_turn(TurnDirection.LEFT)
```

> `add_spiral(...)` 's last two numbers are the **height** then the **width** of the spiral. A direction is `UP` or `DOWN` . A turn is `LEFT` or `RIGHT` .

1 · Predict 🔮

Read each function. Write what you think will happen.

```
def build_hill(line_len):
    rollerCoasterBuilder.add_straight_line(line_len)
    rollerCoasterBuilder.add_ramp(RcbVerticalDirection.UP, line_len)

build_hill(8)
```

The parameter `line_len` is used twice. How long is the straight part, and how tall is the ramp?

```
def build_hill(line_len):
    rollerCoasterBuilder.add_straight_line(line_len)
    rollerCoasterBuilder.add_ramp(RcbVerticalDirection.UP, line_len)

build_hill(4)
build_hill(12)
```

The same function is called twice with different numbers. Are the two hills the same size? Which one is bigger?

```
def build_spiral(spiral_height, spiral_width):
    rollerCoasterBuilder.add_spiral(RcbVerticalDirection.UP, TurnDirection.LEFT,
spiral_height, spiral_width)

build_spiral(10, 3)
```

This function has two parameters. Which number sets how *tall* the spiral is, and which sets how *wide*?

2 · Spot the Bug 🐛

Each block is broken. Read what it should do, rewrite it so it works, then explain why the original was wrong.

Bug A — The code should build a hill 8 long. But when you run it, **nothing happens at all**.

```
def build_hill(line_len):
    rollerCoasterBuilder.add_straight_line(line_len)
    rollerCoasterBuilder.add_ramp(RcbVerticalDirection.UP, line_len)
```

Hint: writing `def` only **teaches** the function. Something is missing after it.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The function should build a hill as long as the number you give it. But `build_hill(20)` and `build_hill(3)` build the **same** hill every time.

```
def build_hill(line_len):
    rollerCoasterBuilder.add_straight_line(10)
    rollerCoasterBuilder.add_ramp(RcbVerticalDirection.UP, 10)

build_hill(20)
```

Hint: look at the numbers inside. The function never actually uses `line_len`.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — This coaster should go **up** and then come back **down** so the cart returns to ground level. But it just keeps climbing into the sky and never comes down.

```
def up_and_down(ramp_len):
    rollerCoasterBuilder.add_ramp(RcbVerticalDirection.UP, ramp_len)
    rollerCoasterBuilder.add_ramp(RcbVerticalDirection.UP, ramp_len)

up_and_down(10)
```

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Build the Loop Coaster 🎢

In your homework world, write **one** function called `loop_coaster` that builds a coaster which **loops all the way back to the starting area**. Your function must have these **four parameters**:

```
def loop_coaster(line_len, ramp_len, spiral_height, spiral_width):
```

Your coaster must include, in total:

- **2 ramps** — one going **UP**, one coming back **DOWN**
- **2 spirals** — one going **UP**, one coming back **DOWN**
- enough **turns** that the track curves around and ends back where it started

Here are the building blocks. They use **fixed numbers** and point the **same way** — fix that:

```
rollerCoasterBuilder.add_straight_line(10)
rollerCoasterBuilder.add_ramp(RcbVerticalDirection.UP, 10)
rollerCoasterBuilder.add_spiral(RcbVerticalDirection.UP, TurnDirection.LEFT, 10, 3)
rollerCoasterBuilder.add_turn(TurnDirection.LEFT)
```

To finish the challenge you must:

1. **Swap the numbers for parameters** — replace `10`, `10`, `10`, `3` with `line_len`, `ramp_len`, `spiral_height`, `spiral_width` so the size is controlled by the call.
2. **Flip the directions on the way back** — change the second ramp and the second spiral from `UP` to `DOWN` so the cart returns to ground level.
3. **Add turns** — keep adding `add_turn(TurnDirection.LEFT)` until the track curves around and meets the start. Add or remove a turn until it closes the loop.

Finally, **call your function once** with numbers you choose, for example:

```
loop_coaster(8, 10, 10, 3)
```

Write your finished `loop_coaster` function and the line that calls it:

4 · Show Your Work 📸 🎥

When your coaster loops back to the start, ride it once in your world.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

> Today I built _____.
>
> I built it using this code: _____.
>
> In this code I used _____.
>
> The hardest part was _____.
>
> That part was hard because _____.
>
> The most fun part was _____.
>
> Something new I learned was _____.

Write your lines here, then say them in your video.

Submit ✅

Send this worksheet + your video to teacher on KakaoTalk.